

MACO v2.2

Multi-Agent Clinical Orchestration with Competitive Logic

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"Clinical safety emerges from structured conflict between constrained expert domains."

Abstract

MACO (Multi-Agent Clinical Orchestration) is a constraint-driven clinical AI framework designed to address safety, explainability, and reasoning transparency limitations commonly found in monolithic large language models used in healthcare settings. Rather than relying on a single probabilistic model to generate end-to-end medical decisions, MACO decomposes clinical reasoning into specialized medical domains represented by independent agents. These agents operate under encoded clinical guidelines and curated domain constraints. The framework introduces a conflict-oriented orchestration model in which disagreements between medical domains are treated as meaningful safety signals rather than noise. A deterministic Historical Context Agent (HCA) enforces immutable patient constraints derived from EHR/FHIR-compatible records, ensuring that critical risks such as allergies, contraindications, and chronic sensitivities cannot be bypassed by model confidence. MACO combines evidence-weighted scoring, formal conflict matrices, negotiation loops, and emergency interrupt protocols to produce explainable and auditable clinical recommendations. The architecture prioritizes deterministic safety enforcement over unrestricted generative flexibility and proposes a systems-level alternative to black-box medical AI.

Introduction

Modern medical AI systems increasingly rely on large generative models capable of synthesizing clinical recommendations from vast datasets. While these systems demonstrate impressive language and reasoning capabilities, they remain vulnerable to hallucinations, hidden reasoning chains, inconsistent guideline adherence, and unsafe therapeutic recommendations. Traditional monolithic architectures compress multiple medical domains into a single latent reasoning space. This creates significant risks in scenarios involving competing organ priorities, conflicting treatments, and patient-specific constraints. In high-risk clinical environments, probabilistic confidence alone cannot guarantee safety. MACO proposes an alternative architecture centered around constrained multi-agent orchestration. Instead of forcing one model to internalize all medical disciplines simultaneously, MACO distributes reasoning across specialized agents supervised by deterministic safety enforcement layers.

Core Philosophy

The philosophical foundation of MACO is that medical truth often emerges through controlled disagreement between specialized reasoning domains. In conventional systems, disagreement is usually treated as instability or failure. MACO instead treats conflict as an explicit diagnostic mechanism capable of revealing hidden risks. For example, a cardiology recommendation may optimize cardiac stability while unintentionally increasing renal toxicity risk. By externalizing these conflicts, the framework enables transparent risk balancing rather than implicit

compromise. This conflict-oriented paradigm transforms medical reasoning from opaque token prediction into structured clinical negotiation.

Design Principles

- Deterministic Safety over Generative Creativity
- Conflict Discovery instead of Blind Consensus
- Domain Isolation for Risk Reduction
- Evidence-Weighted Decision Logic
- Human-Auditable Reasoning Paths
- Explicit Contraindication Enforcement
- Emergency Latency Optimization

System Architecture

MACO consists of five primary components: 1. Specialized Clinical Agents (SLMs) Independent domain-focused reasoning units such as cardiology, nephrology, endocrinology, and infectious disease agents. 2. Historical Context Agent (HCA) A deterministic validation layer enforcing patient history constraints derived from EHR/FHIR-compatible records. 3. Conflict Matrix Engine A structured mechanism for detecting therapeutic contradictions, organ-system risks, and pharmacological incompatibilities. 4. Orchestrator The orchestration controller responsible for proposal ranking, negotiation loops, escalation logic, and final recommendation synthesis. 5. Emergency Interrupt Layer A low-latency bypass system designed for critical conditions such as sepsis, cardiac arrest, and respiratory failure.

Formal Definitions

Specialized Agent Definition: $A_i = \text{SLM}(D_i, G_i, E_i)$ Where: - D_i represents domain-specific datasets - G_i represents encoded clinical guidelines - E_i represents expert-tuned alignment constraints Conflict Modeling: $\text{Conflict}(i,j) = \text{Distance}(T_i, T_j) + \text{ContraRisk}(T_i, R_j)$ Where: - Distance measures therapeutic variance - ContraRisk measures cross-organ danger exposure Proposal Scoring: $\text{Score}_i = (\alpha \cdot C_i) + (\beta \cdot \text{EvidenceLevel}_i) - (\gamma \cdot \text{RiskPenalty}_i)$ Where: - C_i represents internal confidence - EvidenceLevel represents clinical evidence strength - RiskPenalty aggregates detected biological risks

Safety Model

The Historical Context Agent (HCA) acts as a deterministic veto layer. Unlike generative reasoning components, the HCA does not produce recommendations. Instead, it validates every proposal against immutable patient constraints. Examples include: - Allergies - Chronic renal insufficiency - Hepatic contraindications - Laboratory threshold violations - Medication incompatibilities Formal veto rule: $\text{If } T_i \in H_constraints \rightarrow \text{Score}_i = -\infty$ This ensures that unsafe recommendations cannot survive downstream orchestration regardless of confidence score.

Conflict Negotiation Loop

When the global conflict score exceeds a predefined threshold, MACO initiates a negotiation loop. Example: - Cardiology recommends Drug A - Nephrology rejects Drug A due to renal toxicity - Endocrinology raises glucose instability concerns - HCA detects allergy exposure The orchestrator then: 1. Recalculates evidence-weighted scores 2. Evaluates lower-risk alternatives 3. Requests revised proposals 4. Escalates unresolved conflicts when necessary This process converts hidden clinical contradictions into auditable reasoning pathways.

Emergency Protocols

In critical emergencies, deliberative negotiation may introduce unacceptable latency. MACO therefore includes an interrupt protocol capable of bypassing extended orchestration procedures. Examples: - Septic shock - Cardiac arrest - Acute respiratory collapse Upon detection of critical patterns, the system immediately prioritizes: - Lowest-latency intervention - Highest evidence-supported protocol - Minimal biological risk exposure The framework prioritizes safety over deliberative completeness in life-threatening scenarios.

Simulation Example

Patient Profile: - Chronic kidney disease - Penicillin allergy - Cardiac instability Agent Outputs: - Cardiology Agent recommends Drug A - Nephrology Agent flags nephrotoxicity risk - Infectious Disease Agent suggests antibiotic combination - HCA rejects one proposal due to documented allergy Outcome: The orchestrator identifies a lower-risk alternative preserving cardiac stability while minimizing renal burden and respecting allergy constraints. This example demonstrates MACO's ability to transform conflicting recommendations into transparent clinical reasoning.

Explainability Layer

Every final recommendation generated by MACO is accompanied by: - Originating agent identity - Supporting guideline source - Evidence ranking - Rejected alternatives - Conflict rationale - Constraint enforcement history This enables physicians to audit not only what the system recommended, but also why competing

recommendations were rejected.

Threat Model

MACO is designed to reduce: - Hallucinated medical treatments - Unsafe drug interactions - Hidden reasoning chains - Guideline inconsistency - Latency during emergencies - Unverified therapeutic improvisation The framework assumes that probabilistic language models remain fallible and therefore externalizes safety-critical logic into deterministic architectural layers.

Computational Considerations

Key engineering considerations include: - Parallel agent execution - $O(n^2)$ conflict matrix scaling - Negotiation latency optimization - Emergency bypass prioritization - Constraint caching for rapid validation Future implementations may leverage distributed orchestration infrastructure and federated medical reasoning systems.

Limitations

MACO remains a conceptual and architectural framework and has not yet undergone clinical validation. Current limitations include: - Dependence on curated medical datasets - Requirement for accurate guideline encoding - Scalability challenges in large negotiation graphs - Potential orchestration overhead - Necessity of physician oversight The framework is not intended for autonomous unsupervised diagnosis or treatment deployment.

Future Work

Future research directions include: - Reinforcement-based negotiation optimization - Symbolic medical reasoning integration - Real-time EHR interoperability - Federated hospital deployments - Multimodal diagnostics - Explainable clinical simulation environments - Adaptive evidence ranking systems

Conclusion

MACO v2.2 proposes a shift from monolithic generative medical AI toward structured, conflict-driven clinical orchestration. By combining specialized reasoning agents with deterministic safety enforcement, the framework emphasizes explainability, auditability, and explicit biological risk management. Rather than hiding contradictions inside latent model representations, MACO externalizes conflict as a structured safety signal. This architecture provides a foundation for safer, more transparent, and clinically accountable AI-assisted medical reasoning systems.

References

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Monolithic LLMs vs MACO

Feature	Monolithic LLM	MACO
Explainability	Low	High
Deterministic Safety	Weak	Strong
Conflict Handling	Implicit	Explicit
Auditability	Limited	Structured
Emergency Logic	Absent	Built-in